

Run
Center of Genomic Epidemiology (CGE)
Web-based tools

SpeciesFinder
KmerFinder
MLST
ResFinder

Version: April 2025



**Utrecht
University**

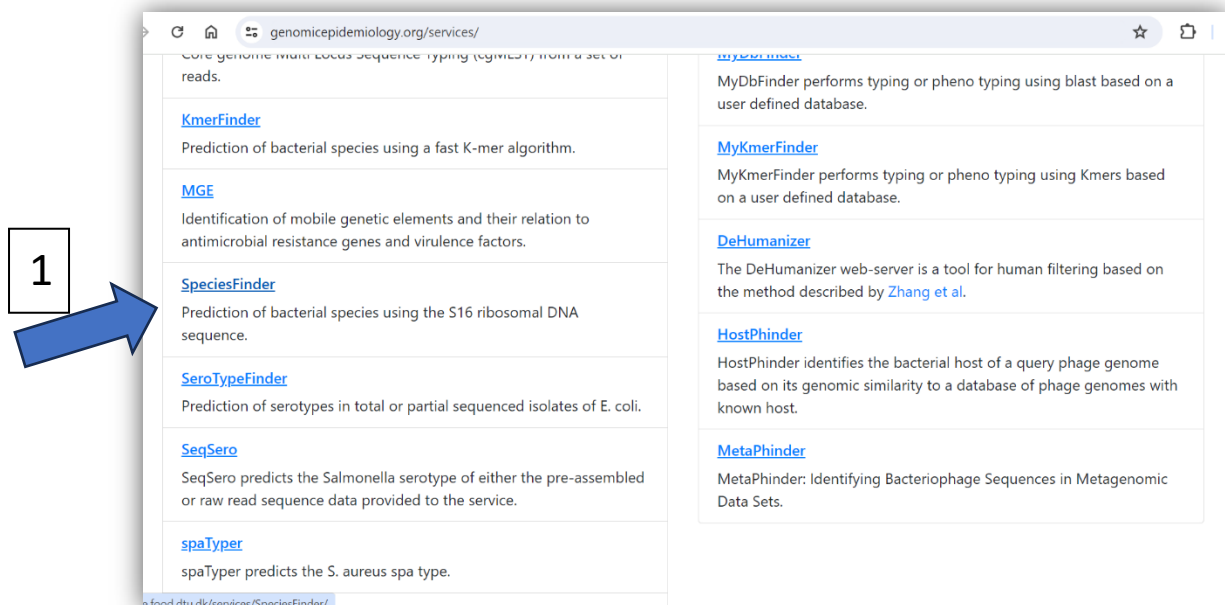
Manual to run CGE webtools

In this manual, we will analyse the genomes with the following tools

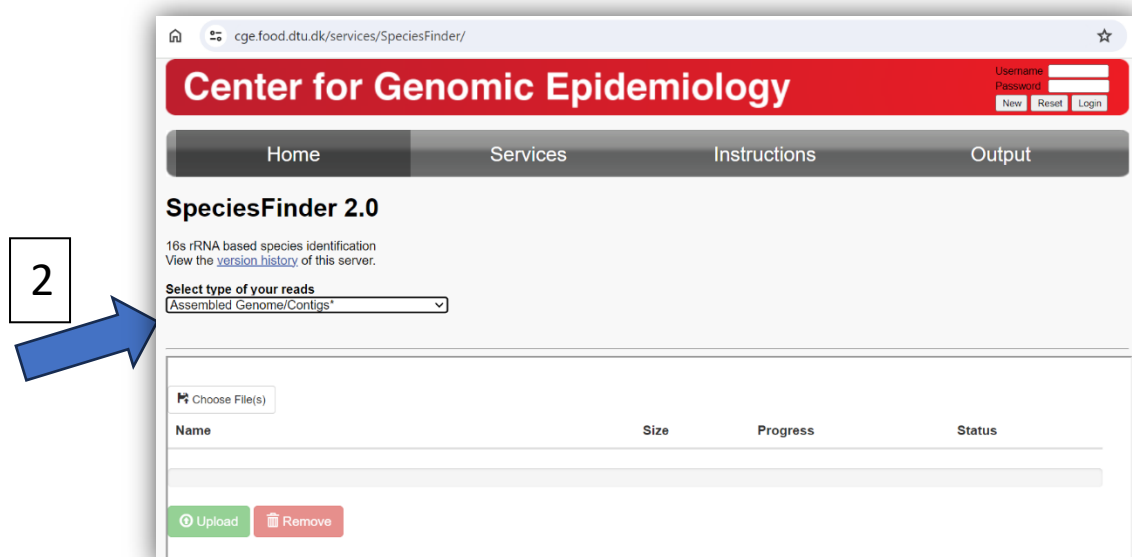
- A. SpeciesFinder
- B. KmerFinder
- C. MLST
- D. ResFinder

Manual SpeciesFinder

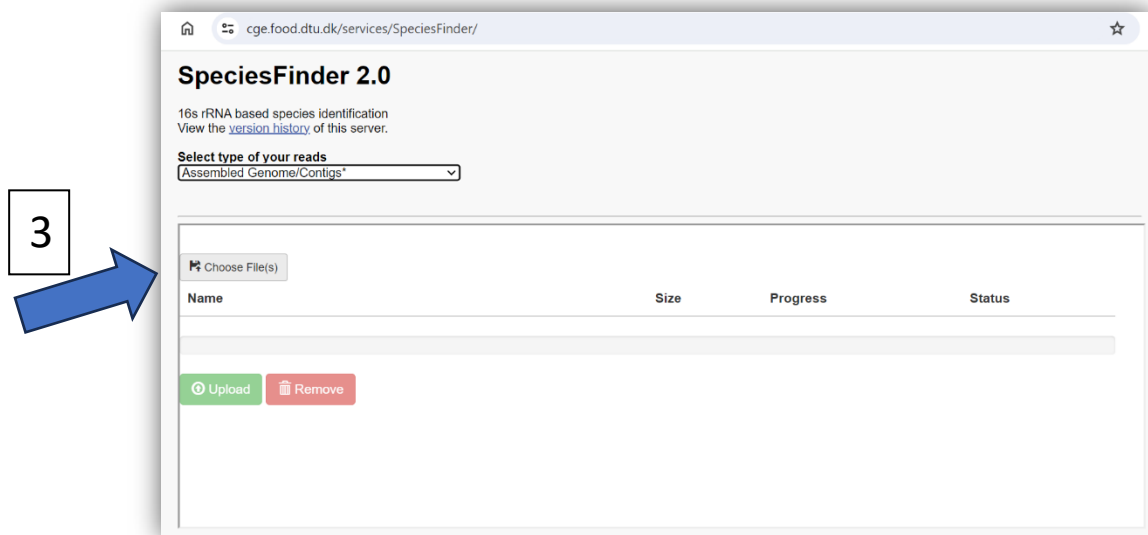
1. Navigate to “Services” webpage of Center for Genomic Epidemiology
<https://www.genomicepidemiology.org/services/>
2. On CGE webpage, navigate to tool “Speciesfinder”



2. On SpeciesFinder page, select type of your reads “Assembled Genomes/Contigs”

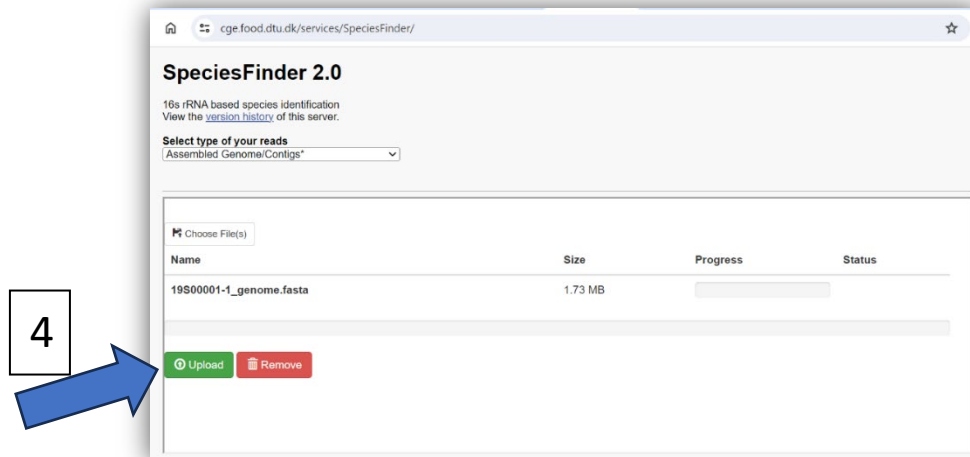


3. Select “Choose File(s)”



4. Browse to downloaded genome file (*.fasta), select file and choose “Open”.

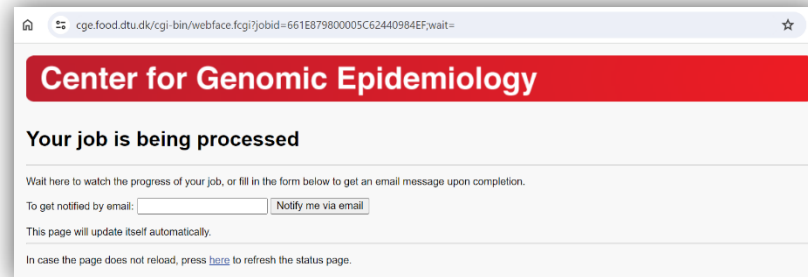
Click “Upload” button.



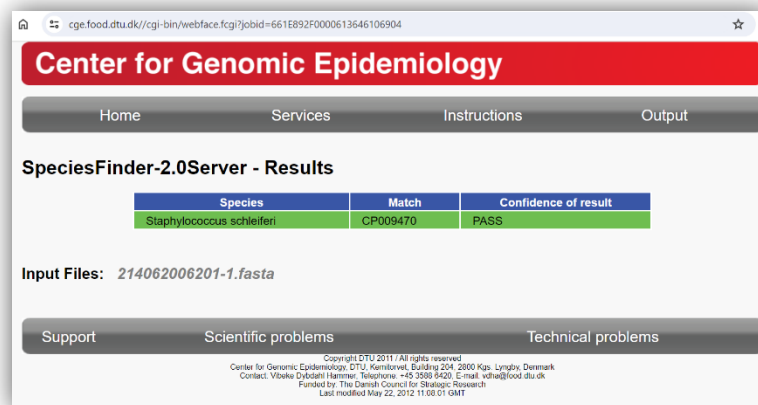
Wait till the progress bar is filled (complete blue)

Page will continue automatically to new page “your job is being processed”

You can fill in your email address to receive an email with the results, or you can wait till the analysis is finished. When the analysis is finished, the page will refresh itself with the results.



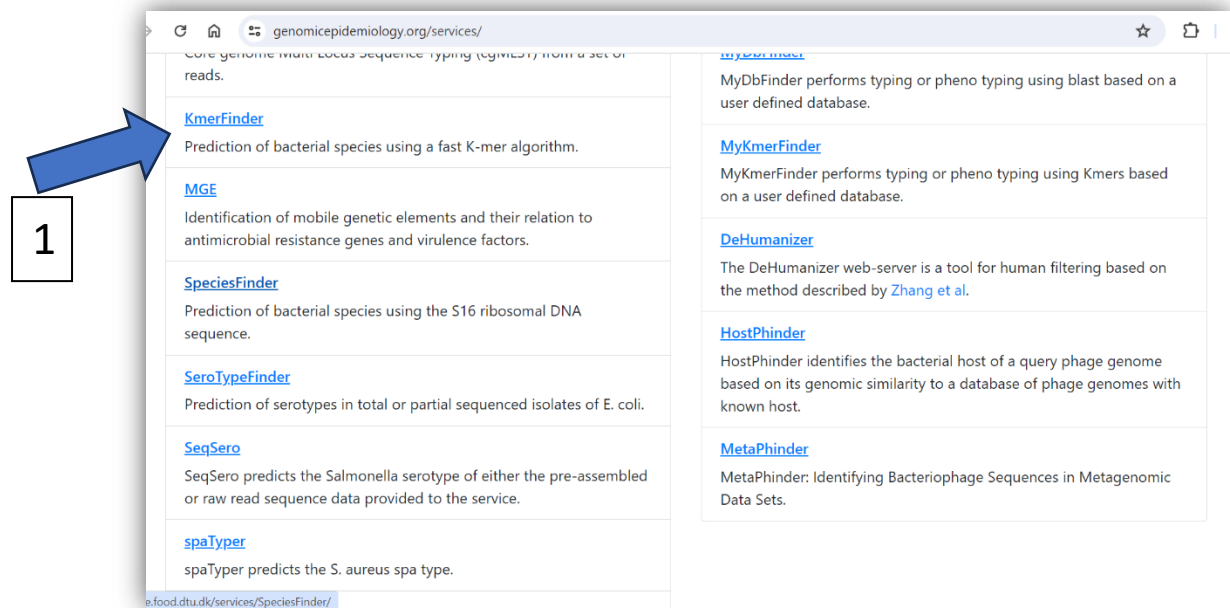
Screen with result is as follows. If the result fail (red), probably your fasta file is not correct uploaded.



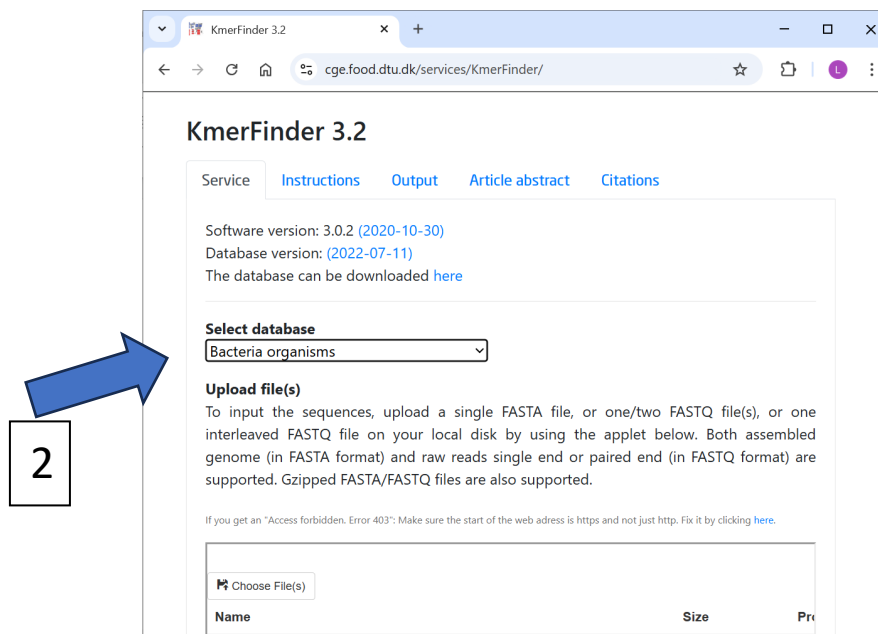
If SpeciesFinder does not give a result; try KmerFinder

Manual KmerFinder

1. On CGE webpage, navigate to tool “KmerFinder”



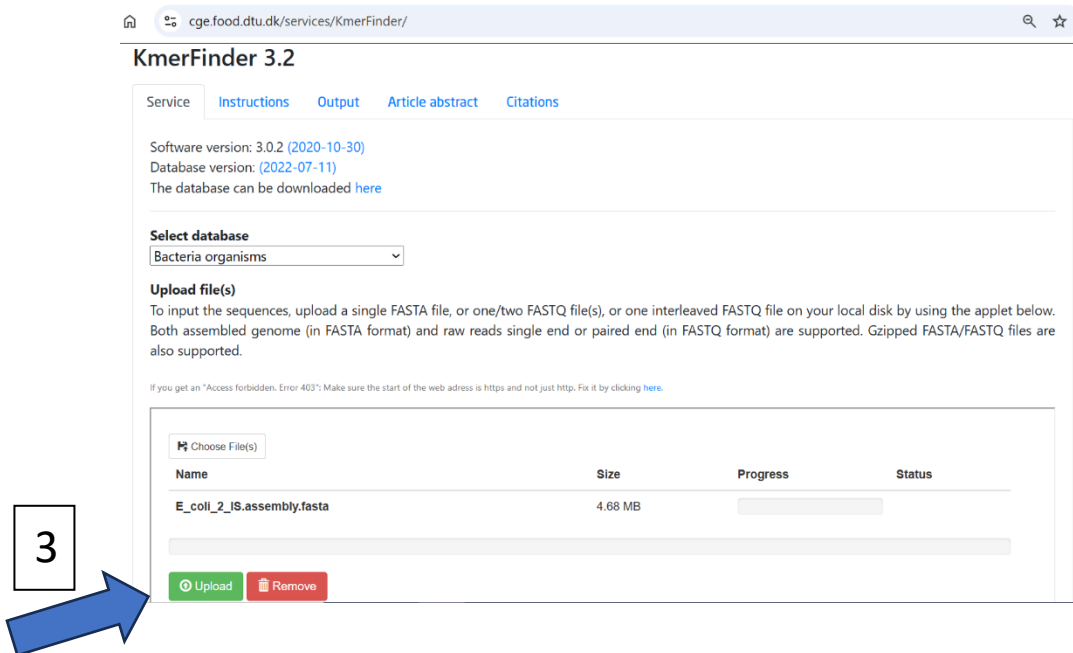
2. On KmerFinder page, select database “Bacteria organisms”



3. Browse to downloaded genome file (*.fasta), select file and choose “Open”.

Click “Upload” button.

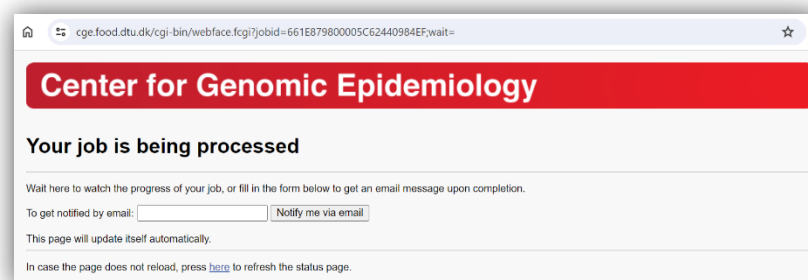
Wait till the progress bar is filled (complete blue)



The screenshot shows the KmerFinder 3.2 web interface. The browser address bar shows 'cge.food.dtu.dk/services/KmerFinder/'. The page has tabs for 'Service', 'Instructions', 'Output', 'Article abstract', and 'Citations'. The 'Service' tab is active. It displays software and database versions, a 'Select database' dropdown menu (set to 'Bacteria organisms'), and an 'Upload file(s)' section. In the upload section, a file 'E_coli_2_IS.assembly.fasta' (4.68 MB) is listed with a progress bar. Below the file list are 'Upload' and 'Remove' buttons. A blue arrow points to the 'Upload' button.

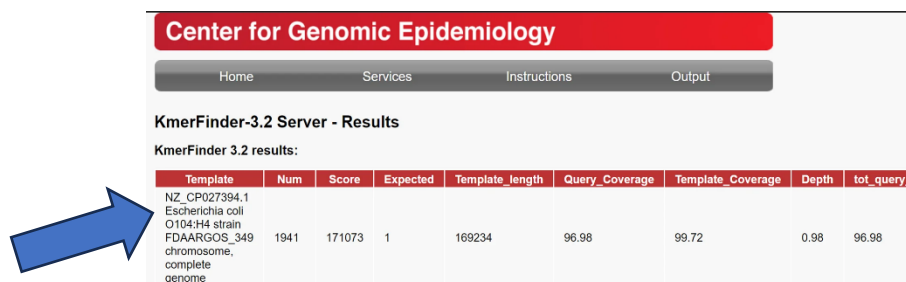
Page will continue automatically to new page “your job is being processed”

You can fill in your email address to receive an email with the results, or you can wait till the analysis is finished. When the analysis is finished, the page will refresh itself with the results.



The screenshot shows the 'Your job is being processed' page. It features a red header with 'Center for Genomic Epidemiology'. Below the header, it says 'Your job is being processed' and 'Wait here to watch the progress of your job, or fill in the form below to get an email message upon completion.' There is a text input field for 'To get notified by email:' and a 'Notify me via email' button. A note at the bottom says 'In case the page does not reload, press [here](#) to refresh the status page.'

Screen with result is as follows. If the result fail (red), probably your fasta file is not correct uploaded.

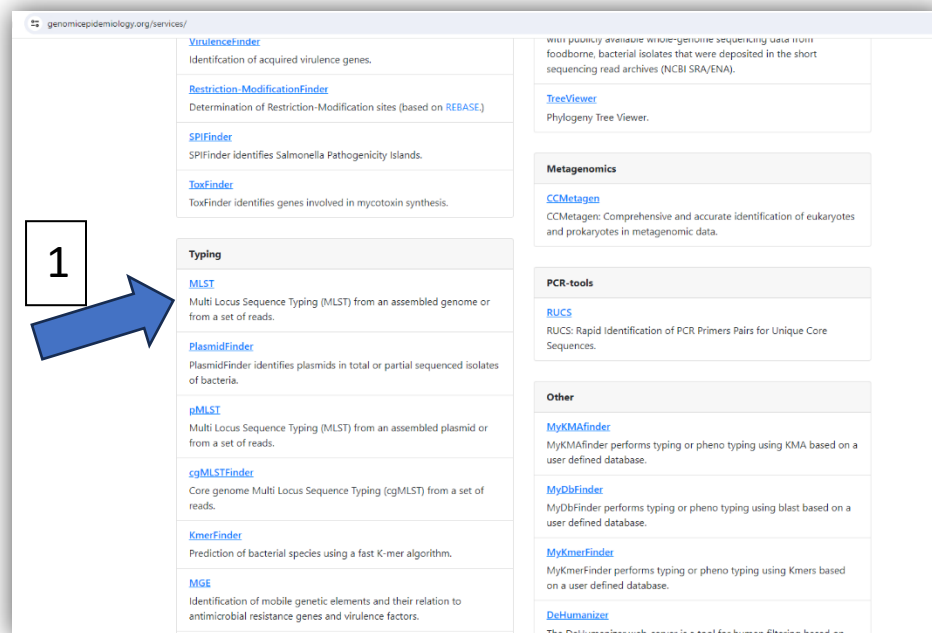


The screenshot shows the 'KmerFinder-3.2 Server - Results' page. It has a red header with 'Center for Genomic Epidemiology' and a navigation bar with 'Home', 'Services', 'Instructions', and 'Output'. The 'Instructions' tab is active. Below the navigation bar, it says 'KmerFinder-3.2 results:'. A table displays the results for the uploaded file.

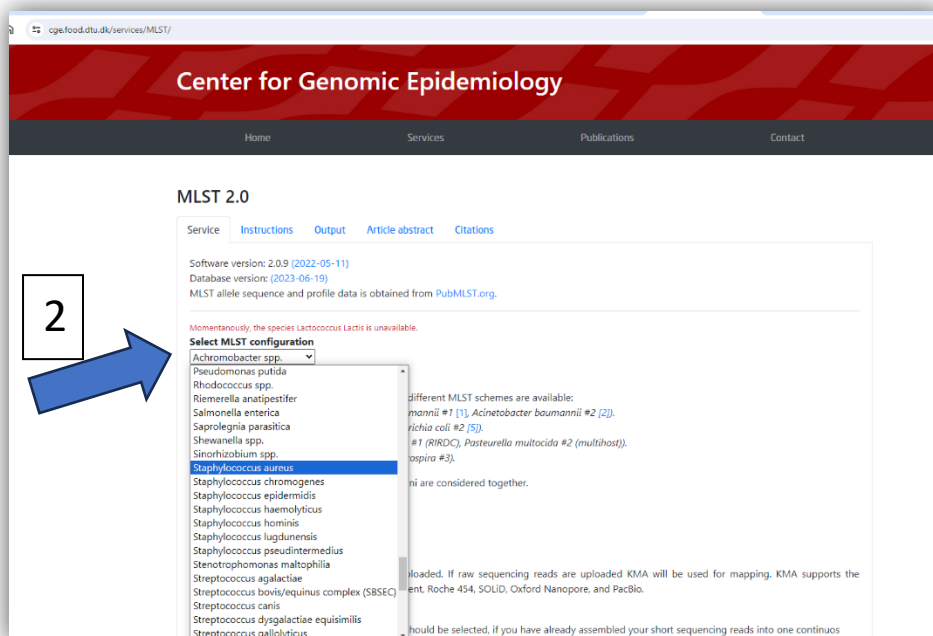
Template	Num	Score	Expected	Template_length	Query_Coverage	Template_Coverage	Depth	tot_query
NZ_CP027394.1 Escherichia coli O104:H4 strain FDAARGOS_349 chromosome, complete genome	1941	171073	1	169234	96.98	99.72	0.98	96.98

Manual MLST

1. On CGE webpage, navigate to tool “MLST”



2. At “select MLST configuration”, select the species of your genome, for example “Staphylococcus aureus”



Use default parameters:

Select min. depth for an allele: 5x

Select type of data input: Assembled or Draft Genome/Contigs

3. Upload your genome file, you know how to do it now
4. Please fill in the Sequence Type (ST) of your genome in the Google docs sheet

Center for Genomic Epidemiology

Home Services Instructions Output

MLST-2.0 Server - Results

mlst Profile: *saureus*
Organism: *Staphylococcus aureus*
Sequence Type: 8

Locus	Identity	Coverage	Alignment Length	Allele Length	Gaps	Allele
arcC	100	100	456	456	0	arcC_3
arcE	100	100	456	456	0	arcE_3
glpF	100	100	465	465	0	glpF_1
gmK	100	100	417	417	0	gmK_1
pta	100	100	474	474	0	pta_4
tpi	100	100	402	402	0	tpi_4
yqjL	100	100	516	516	0	yqjL_3

extended output

Input Files: *sequence.txt*

Please download your results using the buttons below.

[Results as text](#) [Results tsv](#) [Hit in genome sequences](#) [MLST allele sequences](#)

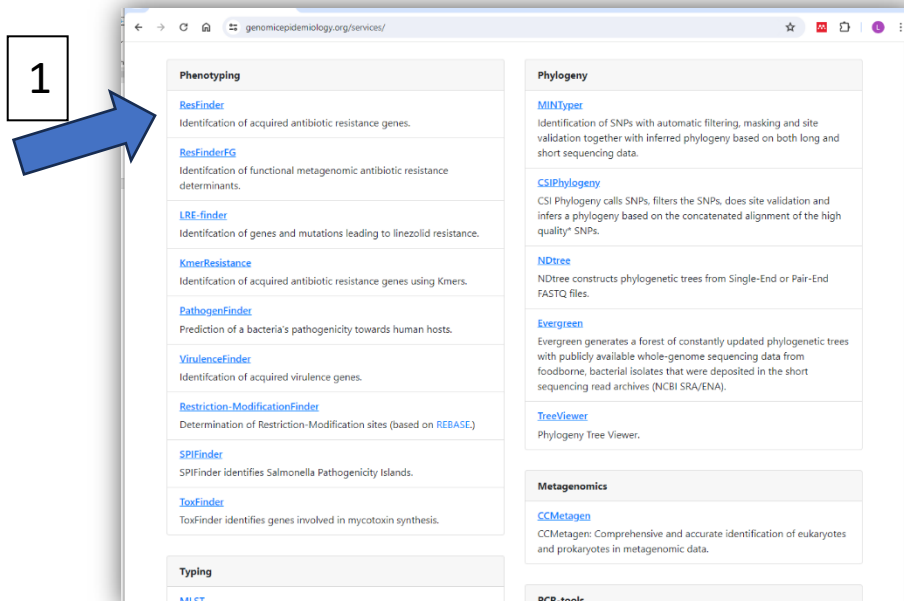
CITATIONS

For publication of results, please cite:

- Multilocus Sequence Typing of Total Genome Sequenced Bacteria.
Larsen MV, Cosentino S, Klemmensen S, Friis C, Hasman H, Mørch RL, Jøsebak L, Scharitz-Pont A, Sn T, Ussery DW, Aarestrup FM and Lund O.
J Clin Microbiol 2012; 50(4): 1352-1361
PMID: [22238442](#) doi: [10.1128/JCM.06094-11](#)

Manual ResFinder

1. On CGE webpage, navigate to tool “ResFinder”



2. On ResFinder page:
 - Chromosomal point mutations:
 - i. Threshold for %ID: 90%
 - ii. Minimum length: 60%
 - Acquired antimicrobial resistance genes:
 - i. Threshold for %ID: 90%
 - ii. Minimum length: 60%
 - Species and input data type:
 - i. Select species of your genome: for example “*Staphylococcus aureus*”
 - ii. Select input type: Fasta (Assembled genome/contigs)
 - Desinfectant: keep switch off
 - Upload and submit job:
 - i. Fill in your emailaddress to get result in your mailbox
 - ii. Choose file: upload your genome

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Services
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ResFinder

Version

4.5.0

ResFinder identifies acquired genes and/or finds chromosomal mutations mediating antimicrobial resistance in total or partial DNA sequence of bacteria.

ResFinder software: (2024-03-22)
ResFinder database: (2024-03-22)
PointFinder database: (2024-03-08)
DisinFinder database: (2023-05-31)

Chromosomal point mutations:

Threshold for %ID

90%

Minimum length

60%

☐ Show unknown mutations
☐ Ignore premature stop codons:
☐ Ignore frameshift indels:

Acquired antimicrobial resistance genes:

Threshold for %ID

90%

Minimum length

60%

2



Species and input data type:
Select species
Other
Select input type
FASTA (Assembled Genome/Contigs)

Disinfectant:
☐ Run disinfectant
Threshold for %ID
90%
Minimum length
60%

Upload and submit job:

Email (Get email, when finished - Optional):

Enter your email address...

Files (The sum of uploaded file sizes cannot exceed 1 gb):

Choose File No file chosen

Choose File No file chosen

Submit Job

Citations

If you use and/or publish results obtained by the service, please cite as described below. If you use raw reads for your analysis, please cite 1 and 2. If you use assembled data for your analysis, please cite 1 and 3.

• Bortolais V, Kaas RS, Ruppe E, Roberts MC, Schwarz S, Cattoir V, Philippot A, Allesoe RL, Rebelo AR, Florensa AR, Fageihauer L, Chakraborty T, Neumann B, Werner G,

3. ResFinder will show which AMR genes are present in your genome, including the predicted phenotype

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Services Contact

Results
ResFinder-4.5.0

Phenotypes Hide

staphylococcus aureus All

Antimicrobial	Class	WGS-predicted phenotype	Genetic background
gentamicin	aminoglycoside	No resistance	
tobramycin	aminoglycoside	No resistance	
amikacin	aminoglycoside	No resistance	
ciprofloxacin	quinolone	Resistant	
ceftioxin	beta-lactam	Resistant	mecA;6;BX571856
penicillin	beta-lactam	No resistance	
sulfamethoxazole	folate pathway antagonist	No resistance	
trimethoprim	folate pathway antagonist	No resistance	
fosfomycin	fosfomycin	No resistance	
vancomycin	glycopeptide	No resistance	
teicoplanin	glycopeptide	No resistance	
clindamycin	lincosamide	No resistance	
quinupristin-dalfopristin	streptogramin a	No resistance	
tiamulin	pleuromutilin	No resistance	
erythromycin	macrolide	No resistance	